TSE*plus*
TSE*fast*

Manual

Juergen Hennig <juergen.hennig@uniklinik-freiburg.de>
Maxim Zaitsev maxim.zaitsev@uniklinik-freiburg.de

1. Overview

1.1. Introduction

TSEplus and TSEfast are based on the writeTSE-sequence from the github example directory of Pulseq (https://github.com/pulseq/pulseq/tree/master/matlab/demoSeq). They do, however, include a number of modifications to allow a wide range of modifications. Both sequences are (nearly) identical, TSEfast includes some features to increase sampling efficiency in long echo train applications like HASTE or SPACE. The difference between TSEplus and TSEfast is outlined in chap.2.4. The following manual applies to both, occasional differences are noted in the respective subchapters, a separate chapter is dedicated to describe the difference between both implementations.
The repository also includes a script TSEreco_prep, which can be used to sort the data into slice-wise rawdata-files for further reconstruction with your favorite reconstruction package (like BART).

DISCLAIMER: The sequence has been gradually developed over the last 3 years. I have somewhat cleaned up the code for consistency, but it is far from being the most elegant implementation. If I had to do it from scratch, I would eg. use the meanwhile available functions to combine and separate gradients much more abundantly. In revising the code, I strictly went by ‘don’t fix it, as long as it works’, so it should not be mistaken as an example on how to properly use Pulseq.

In the following description the struct-names (acqP and acq) of the parameters are sometimes omitted for better readability.

Sequence modes include:

Standard Modes:
- A large range of variable flip angle schemes.
- Different Phase-encoding schemes.
- Acceleration by Half-Fourier and/or GRAPPA/SENSE.
- Flipback to convert residual x-magnetization into z-magnetization to minimize signal saturation effects.
- FatSat
- Inversion recovery

Special Modes:
- Weighted averaging.
- Variable density sampling.
- Navigator echoes in x,y, or z at the beginning and/or end of each echo train.
- An optional prolongation of the first echo time to introduce single spin-echo contrast.

- Diffusion encoding.
- Asymmetric excitation.
- Interleaved acquisition.

All different acquisition modes can be freely combined with each other as far as the resulting sequence is compatible with hardware and/or safety limits of the scanner.

1.2. Sequence structure

Most sequence relevant parameters are stored in structs acqP and acq, where acqP-parameters are those that are usually modified through the regular GUI, whereas acq-parameters are parameters defining the basic sequence parameters like duration of RF-pulses, strength of spoiler gradients etc. The distinction is somewhat arbitrary. Both structs are stored in a parameter-file <seqname>.mat. Per default the seqname is set to TSE_*seqno.* On the first execution *seqno* is set to 1 and from then on incremented by 1 with every execution of the sequence generation script, so while trying out the sequence it may be helpful to reset *seqno* to one every once in a while or to remove the incrementation step.
It is possible to replicate a sequence by loading a previously generated parameter file and setting the parameter setpar to 0.
For generation of multiple sequences with variable parameter sets, the kseq-loop has been implemented. A script TSEplus_para.m is executed at the end of the parameter definition part, whenever nseq>1. Within TSEplus_para.m any of the parameters can be re-defined for every value of kseq to generate multiple sequences in one go. This can be very convenient e.g. for creating series of sequences with different echo times or with different diffusion weighing. But the usage is not limited to that. I do like to keep the basic sequence intact and write whatever features I want to use into TSEplus_para.m. This avoids having to remember all the changes I have made in order to reset the sequence parameter configuration.

A parameter plotflag controls, which plots will be created, depending on which digit of plotflag is set to 1:

#1:	Plot phase encoding ordering
#2:	Plot refocussing flip angles
#3:	Plot k-space components in x,y,z vs. time
#4:	Plot k-space trajectory in x,y
#5:	Plot k-space trajectory in x,z
#6:	Plot sequence diagram.

1.3. Basic considerations for setting up a sequence

Given the large number of parameters for the different acquisition modes, it is advisable to always make a copy of the default sequence and to work with this copy in order to make sure, you always start with a clean slate and don't get lost in parameter space.

Just like the basic writeTSE.m, TSEplus will calculate all relevant sequence parameters from the base parameters set at the beginning of the sequence. Sequence blocks are calculated from these base parameters.

Creating a sequence optimized for a given application requires finding the right compromise between sequence parameters:

- RF pulses should be as short as possible in order to optimize sampling efficiency without violating SAR-limits. Variable Flip Angle schemes can be used to drastically reduce RF poser deposition and thus SAR (s.Appendix A and B).
- The template sequence has two options for RF-pulses: It uses sinc-pulses throughout or an SLR-pulse for the first refocusing pulse in SE-prepared TSE and an adiabatic inversion pulse for inversion recovery. Both of the latter require Sigpy (more about that in Appendix A). They are used, when acq.sigpy is set to 'on'. You can easily put in your own RF-pulses. Timing parameters and gradients are calculated such, that the CPMG-conditions are fulfilled irrespective of the duration of pulses – as long as they are consistent with the chosen echo spacing acqP.TE and the gradient limits.
- Echo spacing: shorter echo spacing allows longer echo trains and therefore shorter overall acquisition time. This comes however at the price of lower SNR due to the higher acquisition bandwidth. Scanning efficiency defined by the ratio of the acquisition time to the echo spacing is also lowered.
- TR should be sufficiently long to incorporate all slices. TR is automatically adapted to the number of slices, so setting acqP.TR to a low value will automatically create the minimum TR for the chosen parameters.
- Readout time should be as long as possible while still allowing sufficient time for gradient rewinders and spoilers.
- Spoiler gradients: The amount of spoiler gradient is determined by the parameters acq.fspR (spoiler in readout direction) and acq.fspS (spoiler in slice selection direction). The spoiler values represent multiples of either the amplitude (acq.spoilermode='amp') or the area (acq.spoilermode='area') of the pertinent gradients – readout gradient prewinder and slice selection rewinder. Default is the latter. A compromise has to be found between the acquisition time and the time left for spoiling. Spoiling in the slice selection direction is more efficient than readout spoiler since slice thickness is typically larger than the pixel resolution. With insufficient spoiling the FID generated by the refocusing pulses will leak into the acquisition window and generate Gibbs-like ringing artifacts.
- No extensive solve handling to resolve parameter conflicts is included except for some timing parameters like TR, TI, TD. In most other cases an error message will most likely occur and it is your responsibility to set up the sequence such, that it is created without conflicts. Typical cases of conflicts are:
- The matrix size acqP.Nx is not compatible with the sampling time acq.samplingTime. acq.samplingTime (in μs) should be an integer multiple of acq.Nx such that the sampling falls onto the 1 μs raster of data acquisition. (In TSEfast the dwell time is set to 1μs and the number of data points is automatically adjusted to the sampling time).
- The sampling time is too long for the given echospacing acqP.TE and the duration of the pulses. Even if the duration nominally fits into the echo spacing, gradient errors may occur, because the residual time may not be sufficient for the necessary spoiler and/or rewinder gradients.

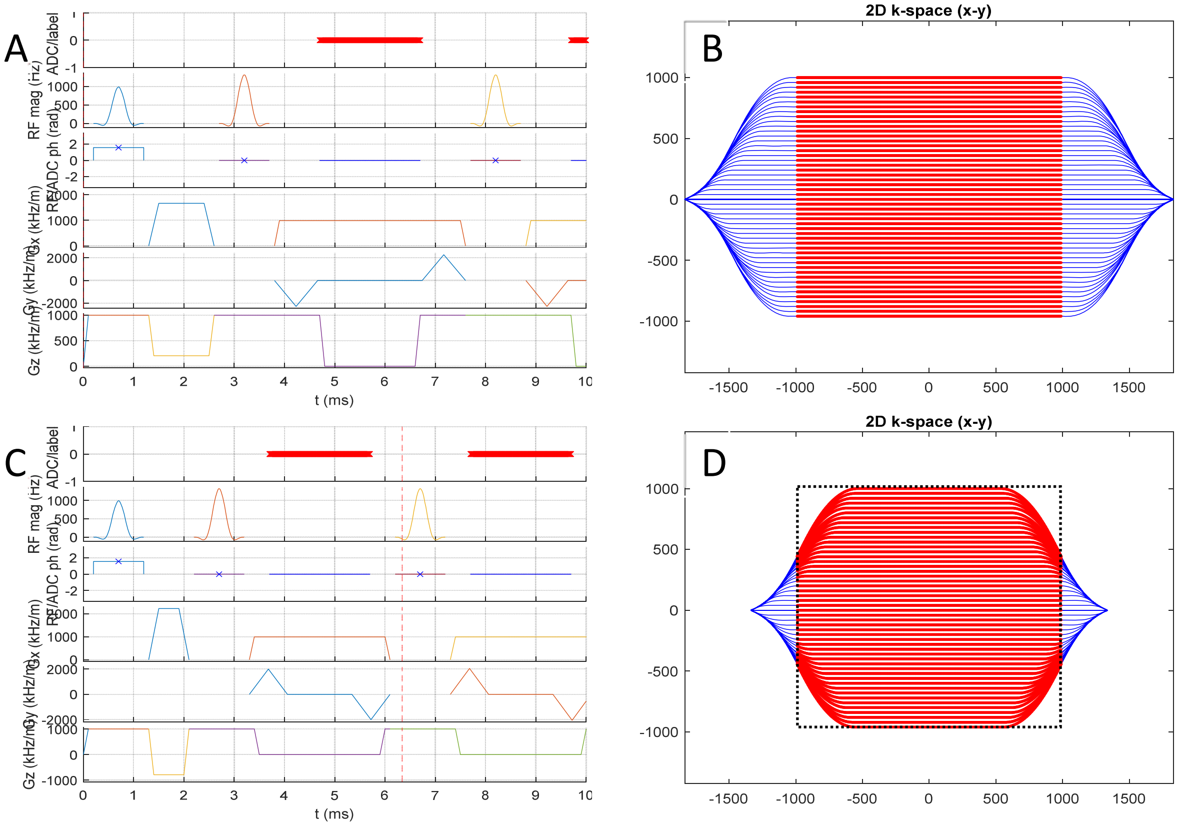
- …

It is advisable to use a dummy scan in order to establish a signal steady state. This is achieved by setting acqP.nDummy to 0 (note the inverse logic).
A warning is produced, if any of the RF-pulses exceeds the maximum B1-amplitude of your scanner. This is set by default to 851.52 = 20 μT, which you may want to adjust according to your hardware specifications and the scanner adjustments.
All imaging gradients have an identical ramp-time defined by acq.dG in TSEplus for the basic imaging gradients except for the prephasing gradient between excitation and the first refocusing pulse in the readout direction, which uses acq.dGs. This may have to be increased in order to avoid PNS-problems. Lowering dG will make the sequence more quiet at the cost of the timing performance and will (slightly) reduce the PNS. For optimal performance dG should be set according to the specs of your gradient system. It should be long enough to accommodate the largest jump in gradient amplitude.
Even if the sequence is created without any error messages, it may still fail on the scanner eg. because of SAR- and/or PNS-errors.

1.4. Difference between TSEplus and TSEfast

The sequence structure of TSEplus follows the conventional approach, in which the echo loop structure is built on four consecutive building blocks: The refocusing pulse is followed by a block of duration tSp, during which phase encoding in Y as well as spoiler gradients in X and Z are applied, followed by data readout and finally another block of duration tSp for the phase encoding rewinder and spoilers. tSp is a dependent parameter and calculated from TE, refocusing pulse duration and readout duration.
TSEfast uses a somewhat different approach and allows an overlap especially between the readout gradient and the phase encoding gradient. This can be used to realize shorter echo spacing for otherwise identical parameters. Phase encoding is performed by varying the duration of the PE-step to minimize the overlap with the readout gradient. The overlap at higher PE-steps leads to a rounding off of the acquired k-space matrix, which requires nonuniform

Fourier transformation for image reconstruction. As shown in Fig.1, the echo spacing is



reduced from 5 to 4 ms while keeping the amplitude and duration of the readout the same. The price to pay is less time for gradient spoilers.

The interleaving of the different gradients is achieved by merging pulses, gradients, and data sampling into a single sequence block making use of mr.addGradients in conjunction with appropriate delays for the individual elements.

2. Sequence Modes

2.1. Standard mode

2.1.1. Variable flip angle schemes

Parameters:

acqP.flipref:	Reference flip angle of refocusing pulses.
acqP.flipflag:	Number of flip angle scheme.
acq.cpmg_mod:	Pulse phase mode

Variable flip angle schemes are essential to realize SAR-optimized sequences . Different flipangle schemes are available determined by the value of acqP.flipflag:

- 0: constant flipangle acqP.flipref
- 1: first refocusing flipangle set to 90+ acqP.flipref/2[1]
- 2: initial flipangles decrease monotonously to acqP.flipref using TRAPS[2].
- 3: as with 0, but decrease to 60° after the echo determined by TEeff
- 4: as with 2, but decrease to 60° after the echo determined by TEeff
- 5: example of a TRAPS-scheme, in which the flip angle is first lowered and then increased to 180° in order to get the full signal at TEeff[3]

The different flip angle schemes are realized with different parameter settings of the function TRAPS_flip.m.

A more extensive discussion of flip angle schemes is found in Appendix B.

acq.cpmg-mod defines the phase scheme for the RF pulses. Per default the phase of the excitation pulse is set to 90° and 0° for all refocusing pulses. acq.cpmg_mod='alt' leads to an alternating phase scheme with 0°(exc)-180°-0°-180°-0°…… acq.cpmg_mod='CP' sets all pulse phases to zero, which will lead to very rapid signal decay but can be used for frequency selective acquisition (s. 2.2.2.).

2.1.2. Phase encoding schemes

Parameters:

acqP.PEtype:	'linear':	linear phase encoding
	'faise':	centric phase encoding according to Melki et al[4]
	'centric':	similar to 'faise' (s.Appendix C).
	'centric_s':	shifted centric.

Linear phase encoding can be safely used, whenever the zero PE-step occurs in the center half of the echo train (25-75% of the echo train length). For shorter TEeff, it is better to use Half-Fourier acquisition (using acqP.HF_fac, s. 1.1.3.) and/or any of the other options. Faise and centric are different, If the number of segments (= number of echo trains) is odd. A more extensive discussion of phase encoding schemes is presented in Appendix C.

2.1.3. Sequence Acceleration

Parameters:

acqP.HF_fac:	Half Fourier factor between 0 (no HF) and 1(skip until reference scans)
acqP.PEref:	Number of reference lines
acqP.GR_fac;	GRAPPA increment

2.1.4. Flipback

Parameter: acqP.flipback

If acqP.flipback is set to 1, a time-reversal of the sequence start (excitation to first refocusing pulse) will be appended at the end of each echo train. With inverted phase of the final 90°-pulse residual transverse magnetization will be converted to z-magnetization. As a result the signal intensity of long-T2-tissues will increase. The effect is stronger at shorter echo train length and short TR. It's efficiency will also depend on the flip angle scheme used.

2.1.5. FatSat

Parameter: acqP.fatsat (on/off)

FatSat is performed using a simple Gauss-pulse with a duration and frequency offset determined by the field strength.

2.1.6. Inversion Recovery

Parameters:
acqP.TI: Inversion time (in s)
acqP.TImod: Inversion mode (0/1)

Inversion recovery is implemented according to the principles of Optimized Interleaved Fluid-Attenuated Inversion Recovery[5], in which the inversion pulse is interleaved with the data acquisition in order to optimally fill the inversion time with echo trains for acquisition. With inversion mode acqP.TImode set to 1, the inversion time is set exactly to acqP.TI, with inversion mode 0, TI may be adjusted to fall in between the echo trains, which may lead to a re-adjustment of the recovery time TR. A more detailed description is given in Appendix D.

2.2. Special Acquisition Modes

2.2.1. Multi-image Acquisition

Parameter: acqP.nrep

In Multi-image Acquisition Mode more than one image is encoded onto each echo train, the encoding pattern of the basic encoding scheme is repeated acqP.nrep times on the same echo train. As an example, data acquisition defined on a sampling scheme with 16 echoes, TEeff=50 ms, TE=10 ms, and acqP.nrep=4 will be realized with an echo train length of 4*16=64 echoes creating four images with TEeff= 50, 210, 370, 530 ms. Images with high TEeff may be of interest for the determination of paravascular spaces in examinations of the glymphatic system and to look for multiexponential T2-decay.

2.2.2. Asymmetric Excitation

Parameter acqP.dTE

In this mode the excitation pulse is shifted by acqP.dTE to the front. As a result off-resonance spins will have developed a phase change at the proper excitation time for the given acqP.dTE. The CPMG-conditions will thus be violated. If the phase shift is 90°, the off-resonance shifts will experience a CP-sequence, which is known to produce a rapidly decaying echo amplitude. This can be used to achieve Fat Saturation without the use of any additional FatSat-preparation. Since this scheme depends on the frequency difference, it does require good field homogeneity (like CHESS- or Dixon-type schemes). It is, however, insensitive to B1-variations and thus especially useful at high field strength. Given a fat-water difference of ~420-440 Hz at 3T, a value of ~0.6-0.7 ms for acqP.dTE will lead to a 90° phase shift for fat.
Using this value and changing the phase of the excitation pulse by 90° will lead to a fat-selective image and destroy all on resonance (water) signals.

Since the amplitude of the first echo is unchanged, it is advisable to use a navigator echo for the first echo (s.2.2.5.). The scheme will work best for longer echo trains, where the k-space center is read out after several refocusing periods. Details are discussed in Appendix D.

2.2.3. Weighted Averaging

Parameters
acqP.wav (0….1);
acqP.wavmode (cont/cat);
acqP.PEinc (0/1)

In weighted averaging mode only the first acquisition is fully sampled, all further averages are performed with an undersampling factor defined by acqP.wav. Fully sampling refers to phase encoding including possible acceleration.
If, for example, acqP.wav ist set to acqP.wav= [1 0.5 0.25 0.25], four averages are performed, but in the second average only the central half of the phase encoding steps are acquired and only a quarter in the 3rd and 4th average. This leads to oversampling of the center of k-space and will improve the SNR compared to two full averages (acqP.wav= [1 1]) at the cost of some blurring. With acqP.wavmode = 'cat', the individual acquisitions are concatenated with each other, with acqP.wavmode = 'cont', the phase encoding steps are sorted to achieve a continuous sampling of PE-steps.
With acqP.PEinc=1 phase encoding in the k-th averaging cycling is increased by 1/nexc, where nexc is the number of averaging cycles (=4 in the example above). This leads to a continuous dataset with (partial) phase encoding oversampling (PEOS). A more thorough discussion is given in Appendix E.

2.2.4. Variable Density Sampling

Parameter: acqP.GRramp (0….1);

Variable density sampling carries the principles of weighted averaging into the readout domain. Rather than acquiring signal under a constant gradient, the readout gradient is V-shaped such, that the echo center is acquired at the lowest gradient amplitude (=low acquisition bandwidth). acqP.GRramp=0 corresponds to a constant gradient, with acqP.GRramp=1 the readout gradient is ramped down to zero at the echo time. Although data sampling occurs at a constant sampling rate, reconstruction has to be performed using non-uniform Fourier Transformation due to the non-equidistant sampling in k-space. Values of acqP.GRramp between 0.2…0.5 yield good results with higher SNR, but without noticeable blurring. A more thorough discussion is given in Appendix F.

2.2.5. Navigator Echoes

Parameter:
acqP.navmode ([0…3   0..3   0…3])
acqP.navk.

Navigator echoes are introduced into the echo train before[6] (first parameter of acqP.navmode) or at the k0th echo interval (second parameter of acqP.navmode), or after (third parameter of acqP.navmode) the echo train used for image encoding. Parameter values indicate:

0: no navigator
1: navigator in readout direction
2: navigator in phase encoding direction
3: navigator in slice selection direction.

Parameter values can be freely chosen, e,g acqP.navmode = [0 0 1] corresponds to no navigator in the first echo, navigator in readout direction at the end of the echo train, [1 0 2] indicates navigator in readout direction before the echo train and in PE-direction at the end.
[0 1..3 0] will lead a navigator echo in the echo number given by acqP.navk. [1 2 0] and acqP.navk=2 will for example produce a navigator echo in read in the first echo and a navigator in the phase encoding direction in the second echo, which can be used for correction of in plane motion in individual echo trains. For navigators in the first echo, the echo train will be shifted by one echo spacing, which needs to be considered in defining TEeff – acqP.navmode=[1..3 0 0] will not allow to set k0=1.
TSEfast only allows navigators in the readout direction for the first echo in SE-TSE mode(s.2.2.6.).

2.2.6. SE-TSE: Prolongation of the first echo spacing

Parameters:
acqP.T2prep (off/on)
acqP.TEprep

SE-TSE uses a first echo spacing, which can be chosen to be longer than the echo spacing of the image readout. This can be used to combine conventional spin-echo-like contrast with a multi-echo readout. SE-TSE will show more pronounced signal attenuation in areas with intravoxel inhomogeneities and less bright fat signal compared to TSE with identical TEeff. The effect is most pronounced for TEprep=TEeff (recommended with acqP.PEmode= 'centric', 'alt', or 'faise'). In the current implementation SE-TSE uses the same pulse shape for the first refocusing pulse as in the echo train. For longer values of TEprep it may be beneficial to use a better shape (like SLR) and/or a refocusing pulse with larger nominal bandwidth to maximize signal. Note, that the beneficial signal retrieval by stimulated echo pathways applies only to the signal after the initial refocusing pulse, the first spin echo will show the $\sin^2(\alpha/2)$-dependency of a conventional spin echo.

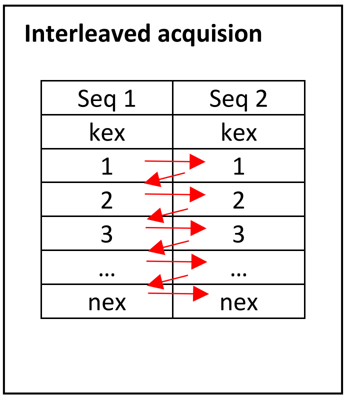
2.2.7. Diffusion encoding

Parameters:

acq.GDs:	Diffusion gradient in slice selection direction (in mT/m)
acq.GDx:	Diffusion gradient in readout direction (in mT/m)
acq.GDy:	Diffusion gradient in phase encoding direction (in mT/m)
acq.tD:	Duration of diffusion gradient in s.
acq.GD:	Ramptime of diffusion gradient

Diffusion encoding is performed by inserting diffusion encoding gradients into the SE-preparation interval. It thus requires SE-TSE-mode (T2prep='on'). Diffusion gradients are inserted symmetrically around the refocusing-pulse into the filling times required for the chosen acqP.TEprep with the maximum possible separation to maximize the b-factor. acq.tD is modified to the maximum duration allowed for the chosen acqP.TEprep, Setting acq.tD to zero will set all diffusion gradients to zero.

2.2.8. Interleaved acquisition



In interleaved acquisition mode data acquisition jumps between two (or more) sequences after each TR. Interleaved acquisition can be used to avoid misregistration errors by motional displacement beween successive scans, which typically occur, when sequences are performed one after the other. In the most straightforward use, TR and the number of slices are kept the same for all sequences eg. for data acquisition with variation of TEeff, dTE or diffusion weighting. The interleaving scheme allows also to interleave between TSEplus and TSEfast or with any other sequence, if it is suitably prepared.
Interleaved acquisition hasn't been fully implemented, but there are markers labeled as 'comment for interleaved acquisition' at those lines in the code, which have to be commented out or in, if this is to be used. The sequence is then reduced to acquire a single TR-loop, which has to be stored as TSEplus_ileave.m. Acquisition is then performed using TSE_ileave.m, in which the sequence is initiated and the kseq- and kex-loops are executed. The different parameters for the interleaved sequences are defined using myTSE_para.
The sequence parameter acqP.nDummy is ignored, since kex is set in TSE_ileave.m, so kex has to be set to run from 0…nex, if dummy scans are to be executed. For Half Fourier and GRAPPA it is advisable to determine nex by running the sequences individually.
Interleaved acquisition has to be used wisely. It avoids misplacement errors, which typically occurs, when the patient moves between sequences. If the total sequence gets too long, patients may get restless leading to intrascan motion and motion artifacts. Too long sequence files my also lead to problems at sequence loading.
The implementation is computationally rather clumsy, for each TR the full sequences are calculated. Since computation time is very short, this isn't relevant in practice. On the plus side this approach enables very fast implementation with a minimum amount of modifications.
Ultimately one would need to store only the relevant parameters for execution. This will come in a future release….

.Alphabetic list of parameters:

acqP-parameters:

Parameter	Chapter	Description
dTE	2.2.2.	Prolongation of the time interval between excitation and first refocusing pulse (in s)
fatsat	2.1.5.	Flag for fat suppression (on/off)

flip	2.1.1.	Actual flip angles applied defined by flipref and flipflag
flipback	2.1.4.	(off/on): Flip residual transverse magnetization back to z-magnetization at the end of the echo train
flipflag	2.1.1.	Flag to define variable flip angle scheme
flipref	2.1.1.	Reference refocusing flip angles
fov		Field-of-view
GDs	2.2.7.	Diffusion gradient in slice direction (in mT/m)
GDx	2.2.7.	Diffusion gradient in readout direction (in mT/m)
GDy	2.2.7.	Diffusion gradient in phase encoding direction (in mT/m)
GR_fac	2.1.3.	GRAPPA undersampling factor (1,2,….)
GRramp	2.2.4.	Ramp factor for variable density sampling. Readout gradient at the center echo is decreased to (1-GRramp) times its nominal value.
HF_fac	2.1.3.	Half Fourier factor (0…1). HF-Fac defines, which portion of the phase encoding lines preceding the reference lines is read out. HF=0: no Half Fourier, all lines are sampled HF=1: sampling starts with reference lines
navmode	2.2.5.	[2..3 1..2 1…3]: Navigator echo in first (first number) or last (second number). Navigator will be in readout (1), phase encoding (2), or slice selection (3) direction.
nDummy		0: dummy scan without phase encoding will be performed-
necho		Number of echoes
nexc*		Total number of excitations performed. nexc=Ny for full sampling and without averaging
nrep	2.2.1.	Number of images encoded on each echo train. TEeff of the images will be TEeff, TEeff+ETL, TEeff+2*ETL,…TEeff+nrep*ETL.
NSlices		Number of slices
Nx		Matrix size in readout direction.
Ny		Base matrix size in phase encoding direction at full sampling
OSlices		Slice order
PEcount		Counter to identify the average number in weighted averaging
PEfac		Factor to relate FOV in phase encoding direction to readout direction. PEfac=1: same fov; PEfac=acqP.Nx/acqP.Ny: same resolution
PEinc		Increment of phase encoding gradient in weighted averaging. PEinc=0: Identical phase encoding steps in all averages. PEinc=1: phase encoding steps are incremented by 1/(number of averages) to achieve monotonous encoding.
PEindex		Index indicating to which average the actual phase encoding steps belong to

PEorder		Matrix of size [number of echoes number of slices] representing the phase encoding order
PEorder0		Matrix of size [number of echoes number of slices] representing the phase encoding order without navigator echoes.
PEover		Phase encoding order goes beyond Ny
PEref	2.1.3.	Number of reference lines for Half Fourier and/or GRAPPA/SENSE
PEtype	2.1.2.	Phase encoding type: 'linear': linear phase encode 'faise': centric phase encoding according to Melki et al() 'centric': similar to 'faise', but with monotonous PE-steps. 'centric_s': centric with horizontal shift of PE-orde.
pow		Relative RF-power compared to full refocusing with 180°
samplingTime		ADC-duration
sat_freq	2.1.5.	Frequency of FatSat pulse
sat_ppm	2.1.5.	Chemical Shift offset for FatSat
sliceGAP		Slice distance as a multiple of the slice thickness (=1: no gap)
sliceThickness		Slice thickness (in m)
T2prep	2.2.6.	Flag for SE-TSE (on/off)
tD	2.2.7.	Length of diffusion encoding pulses (in s)
TE		Echo spacing (in s)
TEeff		Effective TE = echo time for encoding the k-space center. (in s)
TEprep	2.2.7.	Echo spacing of first echo in SE-TSE (in s)
TI	2.1.6.	Inversion time (in s)
TImod	2.1.6.	Inversion mode: 1: Exact TI, TR is adapted if necessary 0: TI is adapted to fit into the time raster defined by TR.
TR		Recovery time
wav	2.2.3.	Vector representing the fraction of the total number of phase encoding steps applied at each average.
wavmode		(cont/cat): in continuous mode the total number of phase encoding steps is resorted to yield smooth encoding scheme.

Acq-Parameters
(* = derived parameters not changeable by user)

accfac		Oversampling factor in read direction
cpmg_mod	2.1.1.	Phase scheme of RF-pulses
ExApo		Apodization of excitation pulse (only TSEfast)

dG		Ramptime of imaging gradients
dGs		Ramptime of prephase gradient in read before 1st refocusing pulse
dGD	2.2.7.	Ramptime of diffusion gradients
extime*		Time available for gradients between excitation and first refocusing pulse
fspR	1.3.	Amplitude of spoiler gradient in the readout direction in muliples of the amplitude or area of the readout gradient during acquisition
fspS	1.3.	Amplitude of spoiler gradient in the slice selection direction in muliples of the amplitude or area of the slice selection gradient during acquisition. The spoiler is added to the gradient necessary for rewinding of the slice selection gradient.
GSex*		Nominal gradient under refocusing pulse.
GSexFac		Attenuation factor of slice selection gradient used to modify slice thickness of excitation pulse
GSIRFac		Attenuation factor of slice selection gradient used to modify slice thickness of inversion pulse
GSref*		Slice selection gradient amplitude under refocusing pulse
GSrefFac		Attenuation factor of slice selection gradient used to modify slice thickness of refocusing pulse
GSSEfac		Attenuation factor of slice selection gradient used to modify slice thickness of first refocusing pulse in SE-TSE
nPinit		Number of initial echoes in which refocusing pulse rfref0 is used before switching to rfref
phaseAreas*		Matrix showing the area under the phase encoding gradients for the phase encoding order given by acqP.PEorder.
RefApo		Apodization of refocusing pulse (only TSEfas)
Reftime*		Time available for gradients between refocusing pulses (= echo spacing minus time used by refocusing pulses)
rfTIdur		Duration of Inversion pulse.
rfTItBwP		Time bandwidth product of inversion pulse.
sigpy		Toggle to use SLR- and adiabatic pulses.
spoilermode	1.3.	‘amp’ or ‘area’ for spoiler scaling factor
tBwPex		Time bandwidth product of excitation pulse
tBwPref		Time bandwidth product of refocusing pulse
tEx		Excitation pulse duration
tRef		Refocusing pulse duration
tRefSE		Duration of first refocusing pulse in SE-TSE

tSpR*	1.3.	Duration of spoiler gradient in readout-direction (only TSEfast).
tSpS	1.3.	Duration of spoiler gradient in slice-direction (for TSEplus tSpR=tSpS).

Appendix A

RF-Pulses

In TSE-sequences RF pulses should be as short as possible in order to maximize the readout time and thus the sampling efficiency. Shorter pulses lead to higher RF-power, so a compromise has to be made. Pulseq allows to generate pulses with different pulse profiles. SLR-pulses yield better pulse profiles, but at the cost of higher RF amplitude and power compared to sinc-pulses, therefore TSEplus uses mainly sinc-pulses for excitation and refocusing except for the refocusing pulse used for spin-echo preparation, which optionally uses SLR-pulses[7]. The inversion pulse used for inversion recovery uses an adiabatic pulse (Note that sigpy should be installed if SLR- or adiabatic pulses are used).

The nominal pulse bandwidth tBw is given by the ratio of the timeBandwidthProduct tBwP with the pulse duration tP. The gradient amplitude is calculated according to the nominal bandwidth and is given by tBw/tP in spatial frequency units. The gradient amplitude for selection of a slice thickness of 4 mm with a pulse with a nominal bandwidth of 1 kHz will be 250 000.

The pulse waveform <pulse>.signal represents the B1-amplitude in frequency units. The time integral over <pulse>.signal therefore directly represents the flipangle (in frequency units). For a pulse with a nominal flip angle of 90° therefore the time integral is always 0.25 irrespective of the shape of the pulse (the nominal flip angle is true only for on-resonance spins).Pulses use apodization in order to minimize sidebands. Typically apodization is set to 0.5, lower values lead to increased sidebands, which induce slice cross-talk. Apodization 0 will produce a rectangular pulse. It should be noted, that the actual bandwidth of the generated response profile is not identical to the nominal bandwidth specified by the pulse duration and the tBwP. As shown in Fig.A1a, for excitation pulses the actual bandwidth approaches the nominal bandwidth only for high values for tBwP i.e. sincs with many sidelobes. For low values of tBwP the response bandwidth shows a strong increase and is more and more determined by the Hanning-filter used for apodization than the specified sinc-profile. Fig.A1c shows pulse shapes and exciation profiles for different values of tBwP and apodization.

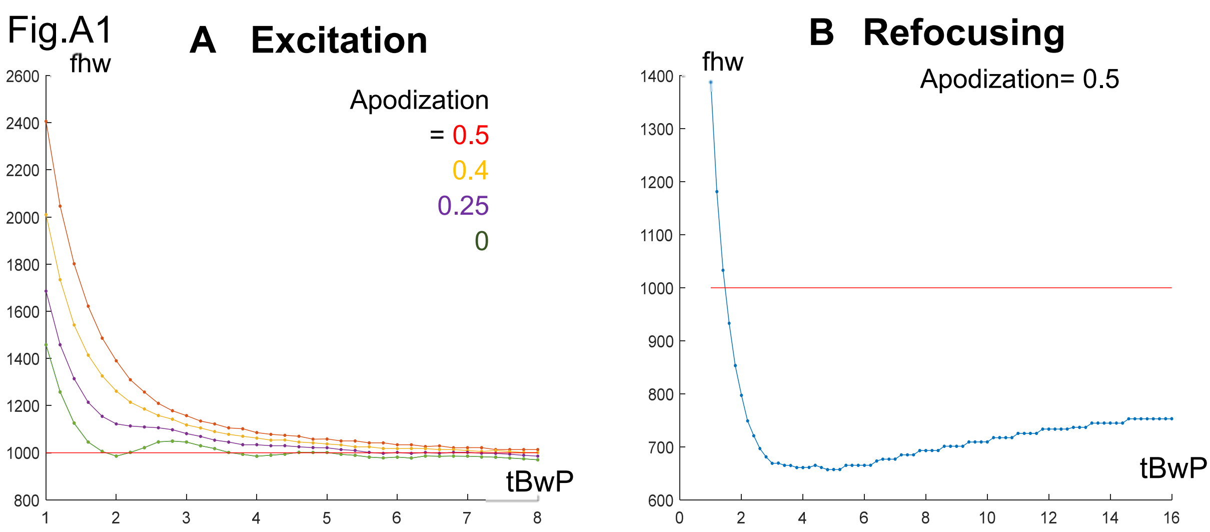


Fig.A1 Half-width of the response profile of excitation(A) and refocusing pulses(B)

For refocusing pulses the refocusing profile shows a much lower bandwidth compared to the nominal bandwidth (Fig.A1b). You need to be aware, that the actual response profiles of excitation and refocusing pulses are quite different even for nominally identical parameters. The gradient amplitude for a given slice thickness is calculated based on the nominal bandwidth, so the actual excitation slice thickness will be higher than specified, whereas the refocusing slice thickness will be lower.

Fig.A2

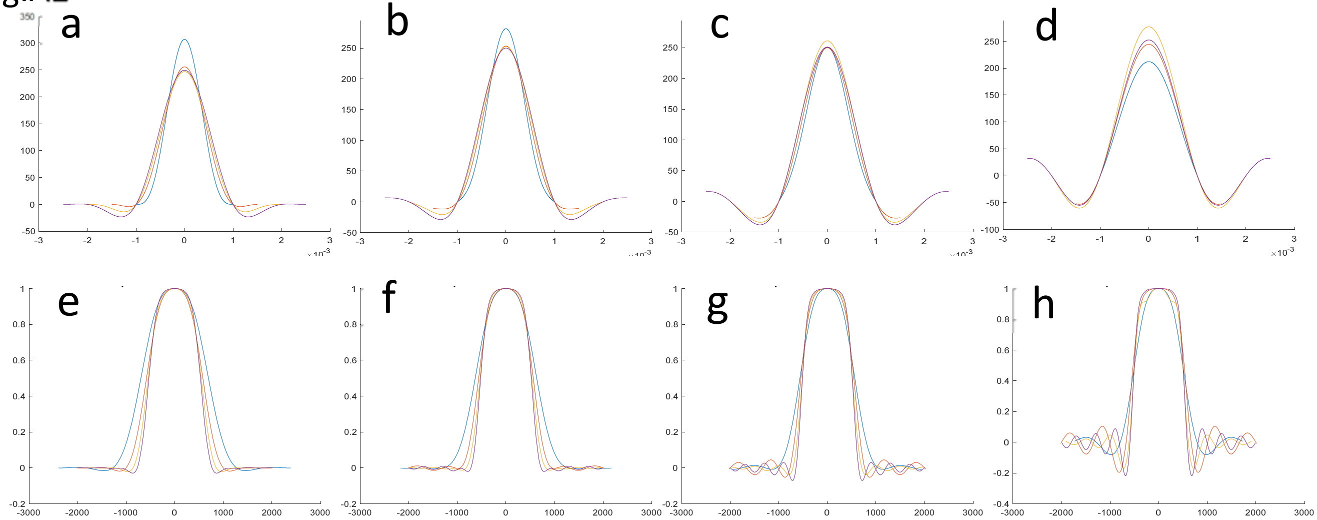


Fig.A2 Pulse shapes (amplitude vs. frequency) of excitation pulses (a-d) for different tPwP and apodization 0.5(a), 0.4(b), 0.25(c) and 0(d). The corresponding response profiles are shown in (e-h). The tBwP is represented by color: 2(blue), 3(red), 4(orange), and 5(magenta).

The refocusing pulse shapes evolve along the echo train due to the superposition of the various refocusing pathways. Due to the stimulated echo mechanism already the second echo shows a pronounced improvement of the signal profile with a somewhat thicker slice and a steeper profile. The refocusing bandwidth increases gradually along the echo train as shown in Fig.A3

Fig.A3

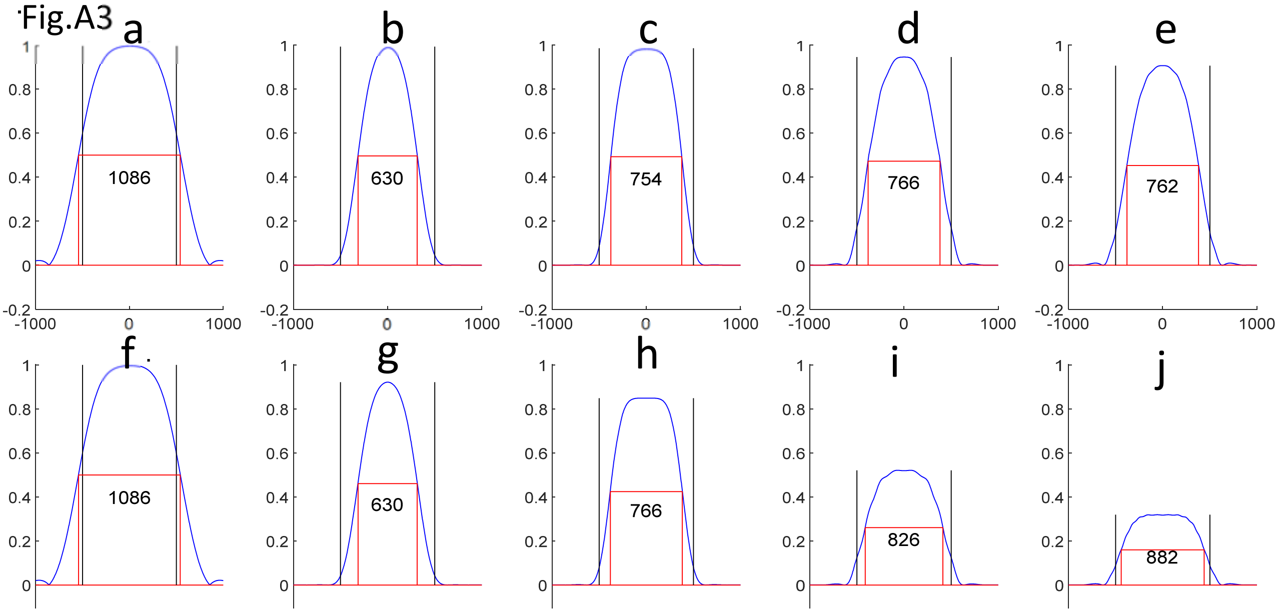


Fig.A3 response profile of the excitation pulse (a,f) and of the first(b,g), 2nd(c,h), 8th(d, i) and 14th(e,j) echo for nominal refocusing flip angle 180° and T1/T2 = 3.65/1.43 ms (CSF, a-e) and 907/92 ms(gray matter, f-j). Excitation pulse duration/tBwP = 4 ms/4, refocusing pulse 2 ms/2;

The increase is more pronounced in tissues with T2<<T1, but even in CSF the effect is observed. The slice thickness thus also will depend on the T1 and T2 of the measured tissues. The refocusing pathways lead to a widening and sharpening of the pulse profiles, even pulses with a tBwP of 2 as shown in Fig.A3 will generate quite reasonable profiles after a few echoes. This allows to use short refocusing pulses with moderately good response profile along the echo trainThe broadening of the response profile along the echo train leads to a somewhat slower signal decay compared to T2 as shown in Fig.A4.

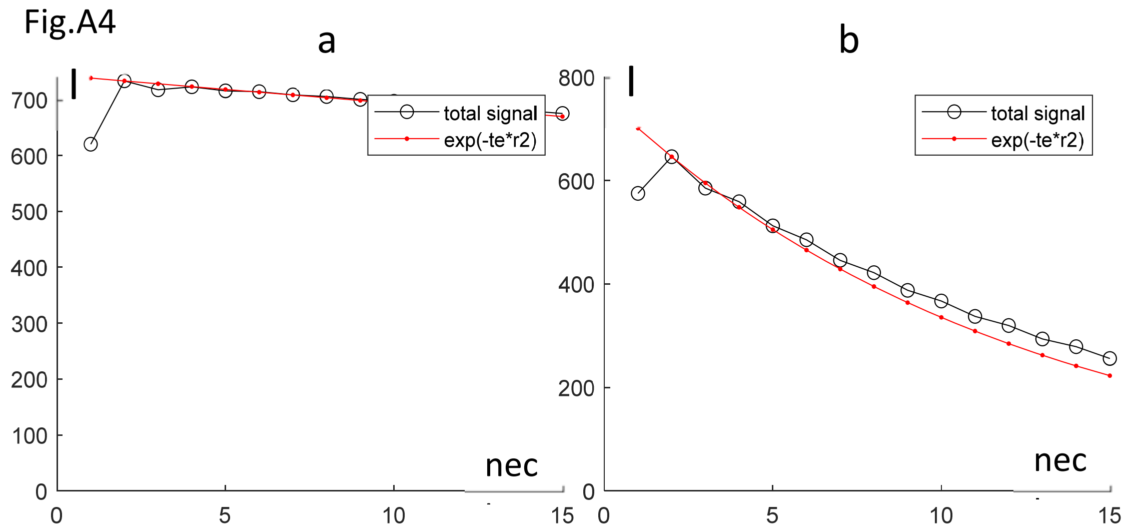


Fig.A4 Signal decay for CSF(a) and gray matter(b) for the parameters used in Fig.A3. Red represents the exponential decay for the respective T2-values.

For the inversion pulse used for inversion recovery measurements it is preferable to use an adiabatic pulse, which has a much better profile compared to a sinc-pulse and the additional benefit of being less sensitive to B1-variations.

The pulse parameters used in the template sequence are by no means optimized, feel free to play around with the parameters.

Appendix B

Flip Angle Schemes

Variable Flip Angle schemes are essential to avoid SAR-problems at field strength of 3T and higher[2,3,8]. The signal intensity can be maximized by keeping the magnetization in the so-called pseudo steady state (PSS). A numerical optimization has been used to make a fast initial transition of the refocusing flip angles towards the target flip angle while maintaining the optimum PSS. The PSS is maintained as long as the change between one refocusing flip angle and the next is not too large (~10-15°).

The predefined flip angle schemes specified in 2.1.1. represent some practical suggestions for some typical use cases.

Fig.B1 illustrates the different flip angle schemes

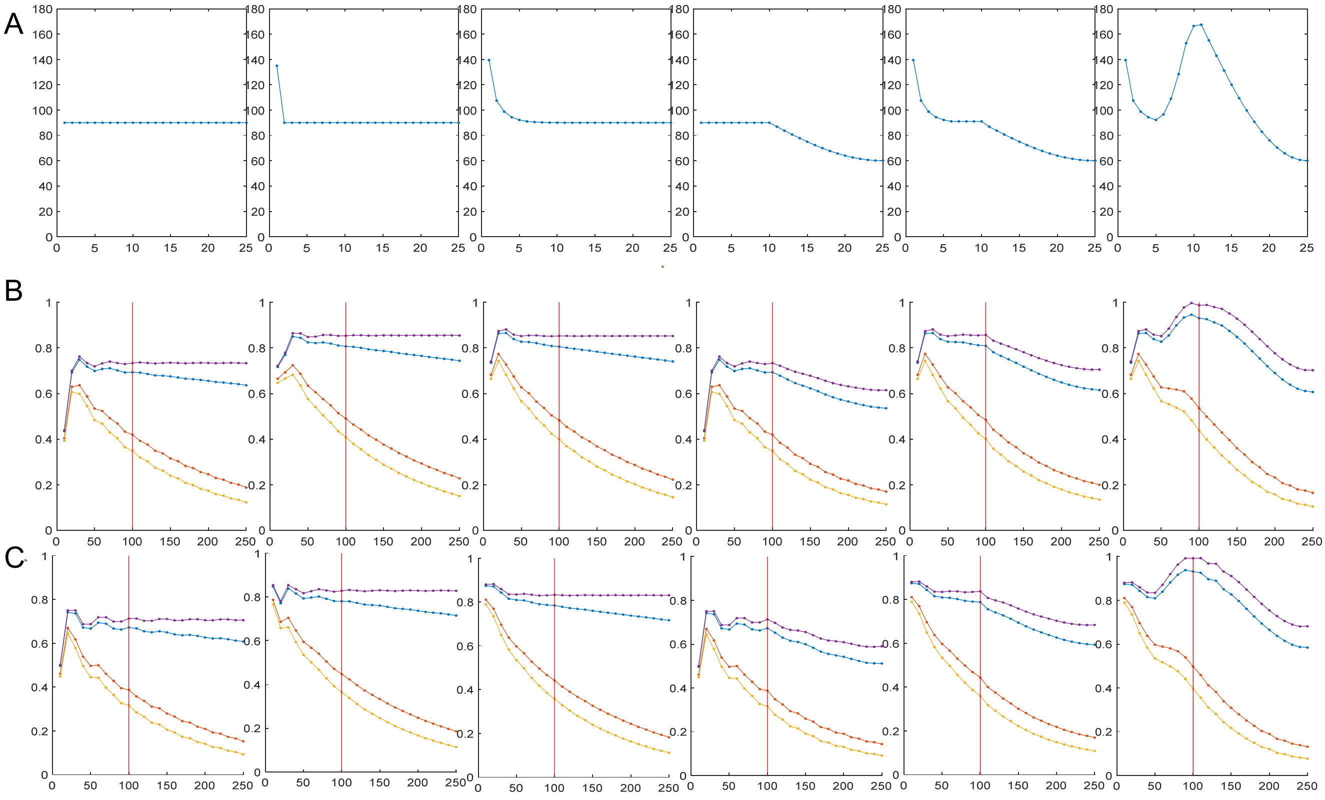


Fig.B1

A) Flipangle schemes for flipflag=0…6 (flipref=90°)

B) Simulation of tissue decay for (from top to bottom: no relaxation(magenta), CSF(blue), gray matter(gm, red), white matter(wm, orange). Simulation used sinc-pulses with tBwP=4 for both excitation and refocusing pulses.

C) Simulation of signal decay using the extended phase graph algorithm with pure flipangles.

Simulations used literature values for T1 and T2 at 3T[9] were used for the simulations. Results for simulations with realistic pulse shapes somewhat deviate from EPG-simulations with pure flip angles, the deviation will depend on the actual pulse shapes used. The most notable difference is the lower intensity of the first echo as discussed in Appendix A. Relative scaling of plots in B) and C) is somewhat arbitrary, scaling to the highest intensity of the signal with no relaation was used.

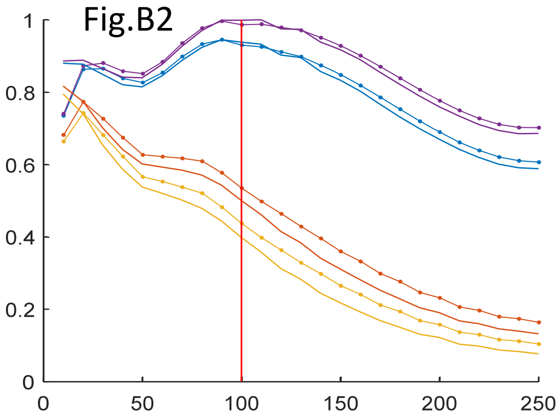


Fig.B2 shows an overlay of the plots based on EPG-simulations (colored lines) with simulations using realistic pulse shapes (colored lines with dots). For no relaxation and CSF the agreement is excellent, for gray matter and white matter intensities for realistic pulses are somewhat higher due to the broadening of the pulse profiles as discussed in Appendix A. The vertical line indicates the echotime at which the k-space center is acquired.

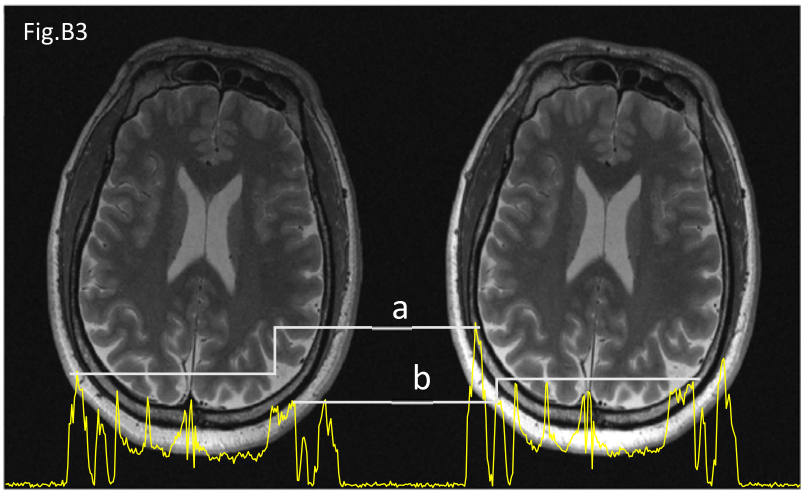
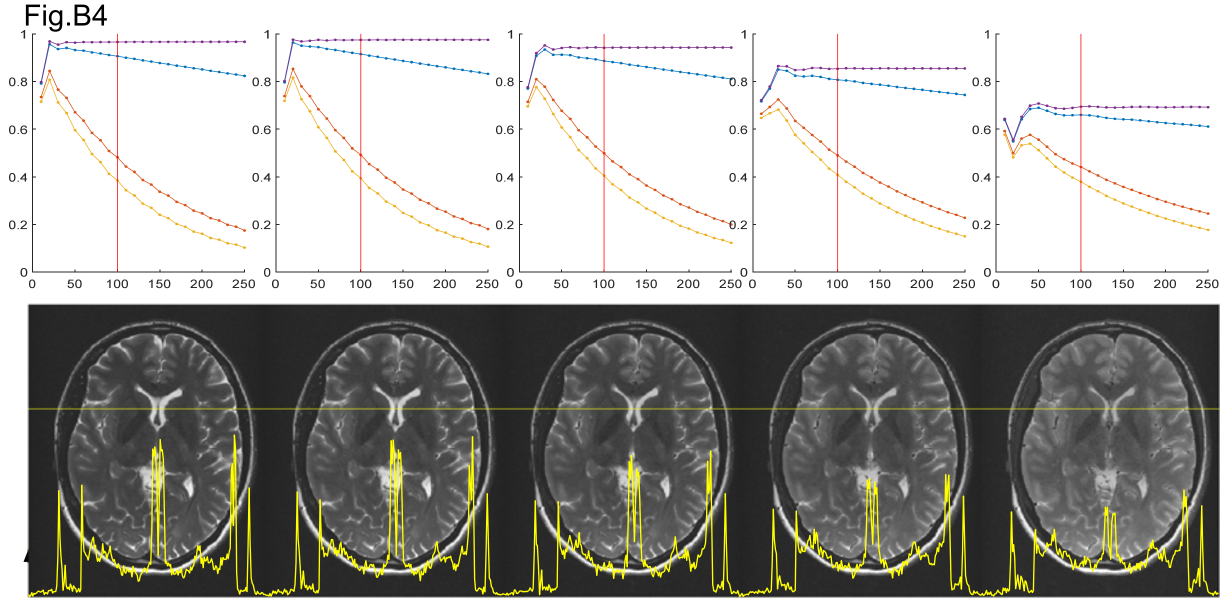


Fig.B3 shows a comparison of images acquired with flipref=60° and flipflag=4(left) and 5(right). Signal of fat (a) and subarachnoid CSF (b) is clearly increased in accordance with simulations. The overall improved image quality of the image on the right is clearly demonstrated – which comes at the price of overall higher RF power.

Fig.B4 shows simulated signal intensities and measured images acquired with variable refocusing flipangles (flipref= 180, 150, 120, 90, 60) and flipflag=2. Image appearance is nearly identical for the first three images, at flipref=90° some subtle change is noted, which becomes clearly visible at flipref=60°. As demonstrated by the profiles taken at the horizontal line in the images, the change of appearance is mainly due to the decrease of the brightest features (CSF and fat), signal intensity of gray and white matter is only slightly increased.

Fig.B4



Phase Encoding Schemes

Different phase encoding schemes are available. The most commonly used are linear and centric. In the centric phase encoding scheme phase encoding order goes in a wedge-like fashion from the center of k-space out in the positive and negative direction respectively (s.Fig.C1A). For an even number of echo trains the positive and negative branch are acquired in successive scans (Fig.C1B). There are two options to place the k-space center k0 into a later echo: to either shift and permute the wedge to the right in 'centric_s' mode as shown in C or in the vertical direction labelled as 'centric' or 'faise'(D). The latter is loosely based on the method proposed by Melki[4] with some modifications: Eq.4a and b in ref.[4] will lead to fractional values for the phase encoding order for odd numbers of echoes and/or segments resulting in slight discontinuities in the k-space coverage. This has been corrected and extended to also allow half-Fourier- and GRAPPA-sampling schemes. The upper row in (C-E) shows the distribution of phase encoding steps along the echo train for different phase encoding schemes, k0 marks the echo in which the k-space center is read out; the middle row demonstrates the exponential signal decay along the echo train vs. the phase encoding order; the bottom row shows the resulting point spread function PSF.

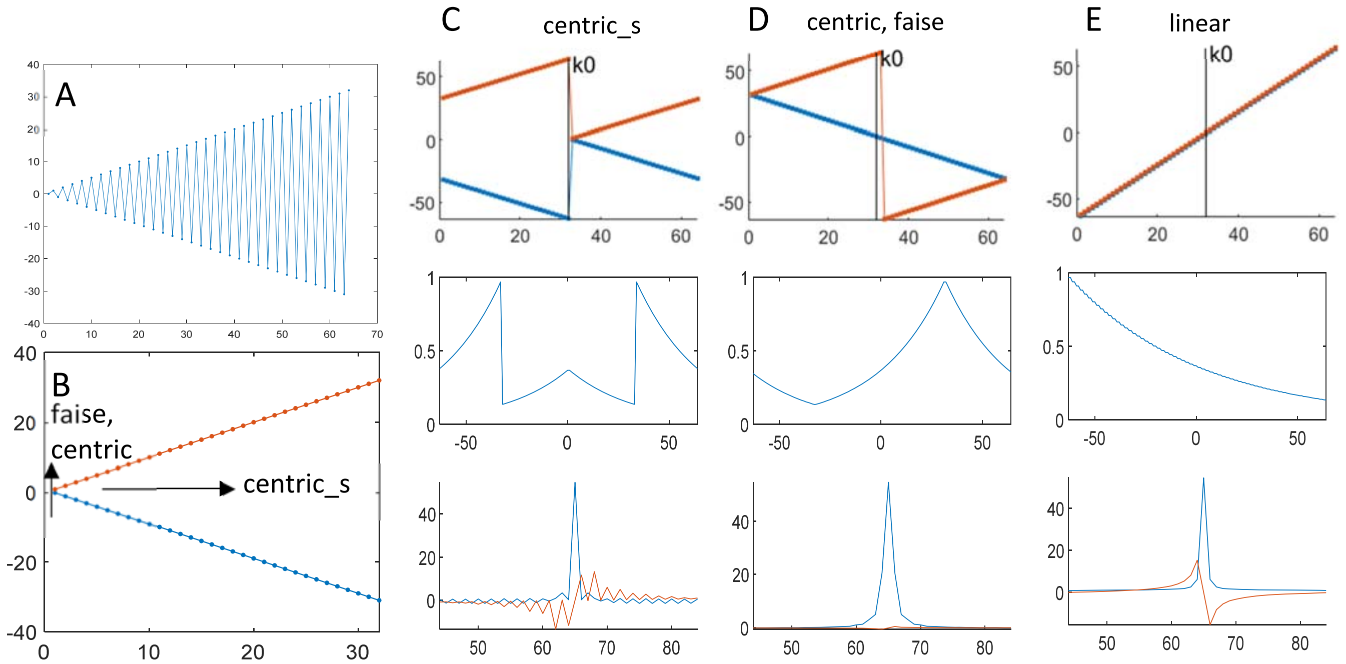


Fig.C1. For explanation see text.

It is clear, that the PSF of the 'centric_s' mode is unfavorable due to the high intensity of the peripheral parts of k-space, whereas 'faise' shows a periodic weighting (D, middle) and thus no sidebands in the CSF. Signal decay across k-space is twice as fast as in linear encoding (E, middle), therefore the PSF of linear encoding is a sharp Lorentzian line but with a non-zero imaginary part (red line in E, bottom) due to the asymmetric FID-like exponential k-space weighting. The PSF can be modulated by normalizing the echo decay.

If k0 is set to occur before the center of the echo train, some wrap-around will occur in linear phase encoding (Fig.C2a). For very small values of k0 – as used in T1-weighted scans – this may lead to jumps of the signal intensity close to the k-space center, which may result in

Fig.C2

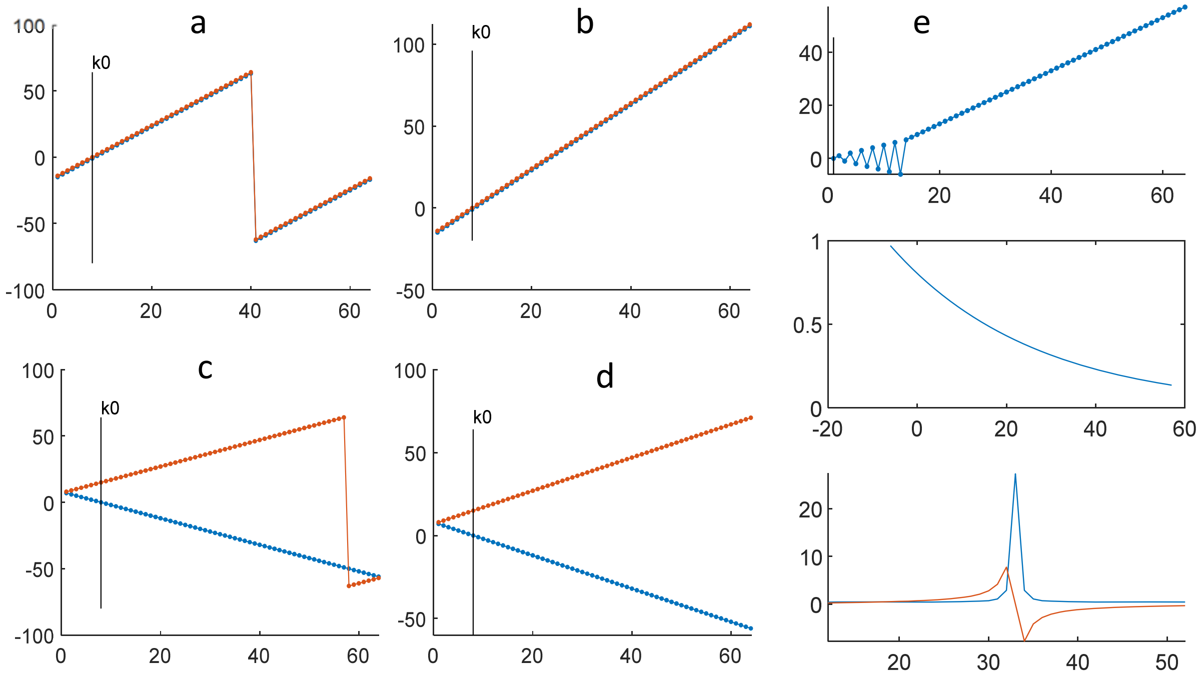


Fig.C2 Linear (a,b) and faise(c,d) phase encoding order for small k0. The jump in (a,c) is removed by setting acqP.PEcont to 1 (b,d). For placing k0 into the first echo it is favorable to use 'centric' phase encoding with a high half-Fourier factor HF_fac (e), where the reference lines for half-Fourier reconstruction are sampled in a centric fashion followed by linear phase encoding of the measured half of k-space.

artifacts. This can be avoided by setting acqP.PEcont to 1 (Fig.C2B). This will lead to a continuation of the PE-order outside of the range set by -Ny/2…0….Ny/2. As a rule of thumb the asymmetric k-space coverage still allows image reconstruction by direct FFT as long as k0 stays in the middle half of the acquired k-space. For smaller values of k0 half-Fourier reconstruction will have to be used. As an alternative approach the echo train length can be shortened and Half-Fourier encoding used directly. acqP.PEcont=1 can also be used for PEtype='faise' (Fig.C2, C and D).

Very early readout of the k-space center can be realized with PEtype ='centric' in connection with a high half-Fourier factor acqP.HF_fac. This even allows to place k0 into the first echo, the reference lines for half-Fourier reconstruction are then read out in centric mode followed by a linear phase encoding order along the rest of the echo train (Fig.C2E). 'centric_s' and 'centric' encoding in this case show a very beneficial and sharp PSF.

The difference between the 'centric' and 'faise' mode lies in the different handling of segmented acquision with an odd number of segments. This is illustrated in Fig.C3, which demonstrates, that 'centric' and 'faise' yield identical phase encoding order for even number of segments, but differ in the handling of odd number of segments.

Fig.C4 shows images acquired with the different phase encoding modes. Images look identical in spite of the quite pronounced apparent differences of the phase encoding ordering.

Fig.C3

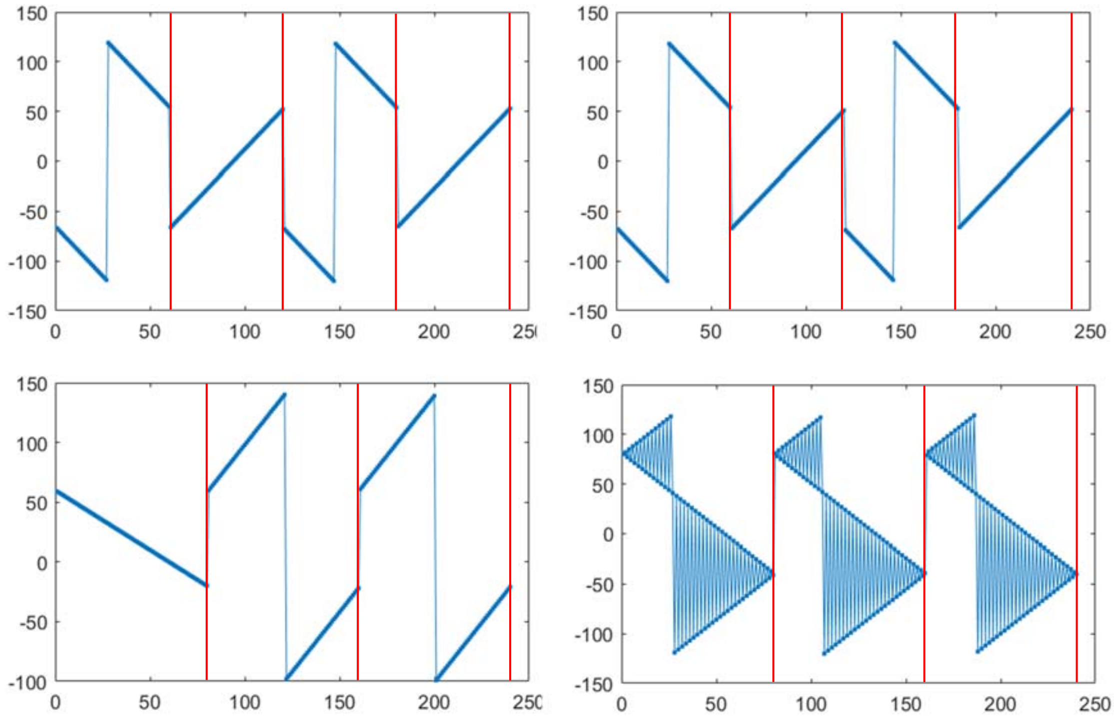
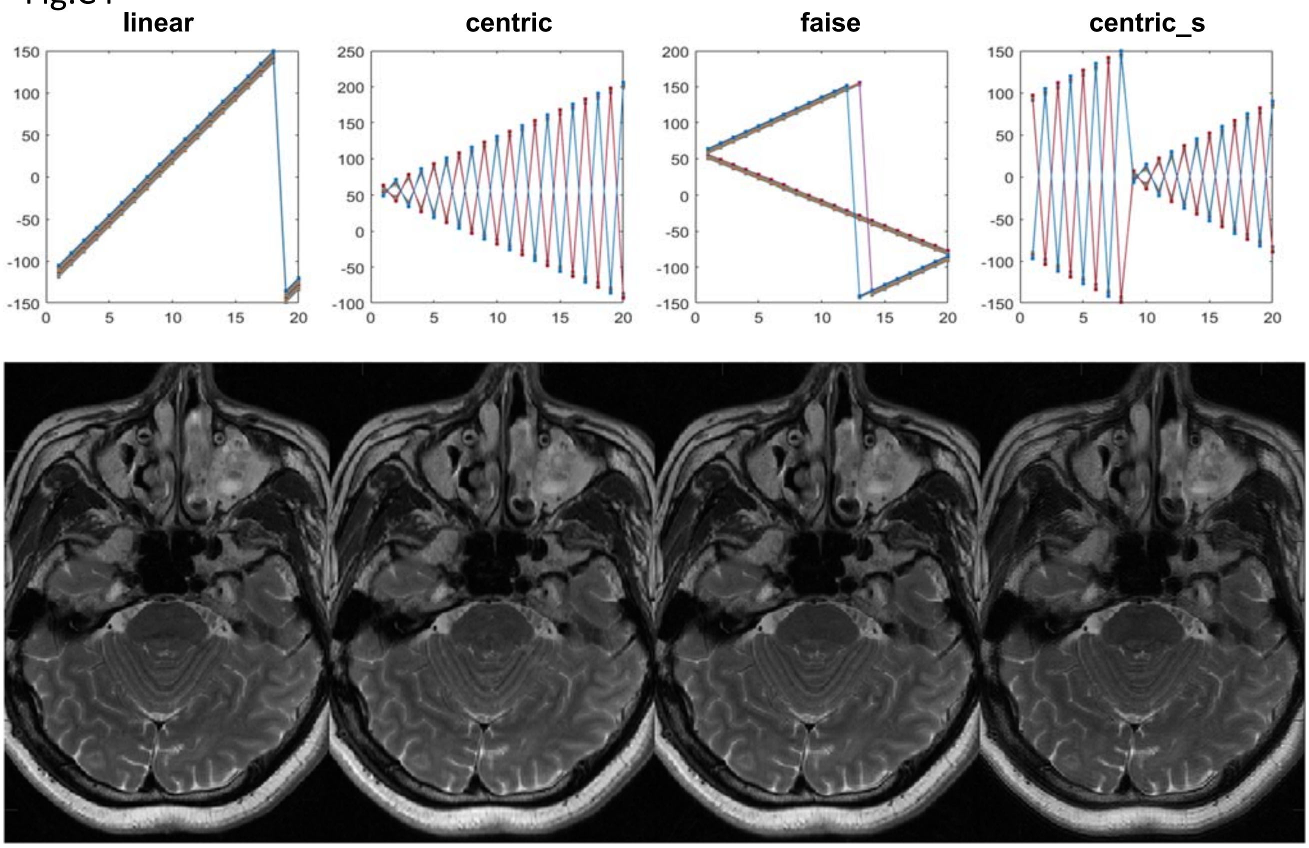


Fig.C3. Phase encoding order for 'faise' (left) and 'centric' for segmented acquisition with 4(top) and 3(bottom) segments. For an even number of segments, both phase encoding schemes yield identical phase encoding ordering, but they differ for odd number of segments.

Fig.C4



Appendix D

Asymmetric Excitation

The CPMG-conditions require, that the phase of the excitation pulse is orthogonal to that of the refocusing pulses. The most simple way to meet this condition is to use phase of +-90° for the excitation pulse and 0° for the refocusing pulses. It should be noted that due to the frequency invariance of the Bloch equations this can be generalized to

$\pm 90° \pm \alpha - \pm 2\alpha - \pm 4\alpha - \ldots . - \pm 2ne\ \alpha$,

where α represents and arbitrary phase offset and ne indicates the echo number. For $\alpha = 90°$ this leads to the commonly used alternating phase scheme

0° - 180° - 0° - 180° - 0° - 180° - 0° -….

CP-conditions leading to destructive interference of the different refocusing pathways occur, if the excitation pulse is in phase phase the refocusing pulse. This is illustrated in Fig.D1 for simulations using sinc-pulses for both the excitation and the refocusing pulses.

This can be used for chemical shift selective signal attenuation eg. in fat suppression. If the time between the excitation pulse and the first refocusing pulse is changed by a time interval dTE, where dTE leads to a phase evolution of 90° for fat signals, then fat will effectively 'see' a CP-sequence with low signal intensity, whereas on-resonance spins show the beneficial CPMG-signal decay. dTE should be $1/(4\Delta f)$ to achieve this condition, which corresponds to ~0.6-0.7 ms at 3T. For $dTE=1/(2\Delta f)$, fat will be out-of-phase to water. Fat-selective imaging is achieved using $dTE=1/(4\Delta f)$ combined with CP-conditions for the pulse phases, fat will then evolve to 'see' CPMG-conditions. Note that the signal in CP-conditions is high at the first echo, so this will not work at short TEeff.

Fig.D2

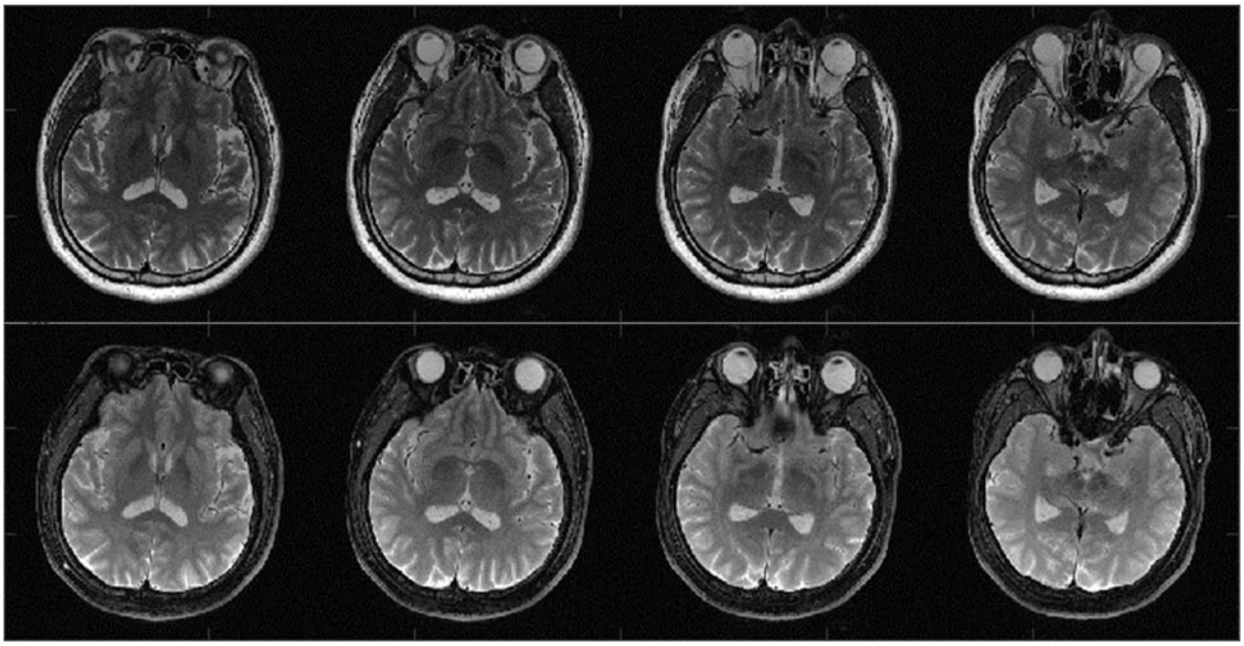


Fig.D2 TSE with asymmetric excitation (bottom row) with dTE=0.7 ms compared to standard TSE with dTE=0 (top row) (TE/TR= 80/4000 ms, ESP=10 ms)

As with other chemical shift-dependent fat suppression schemes, signal in areas of strong local field inhomogeneity is also affected. Fig.D2 shows results of measurements with echo spacing 10 ms, TEeff 80 ms and dTE=0.7 ms. Note the good suppression of orbital fat. Signal above subarachnoidal spaces is also attenuated due to local susceptibility effects..

The intrinsic fat suppression offered by the asymmetric excitation scheme is especially suitable for applications at high fields, where B1-inhomogeneity may not offer good fat suppression by shift selective pulses, it also saves RF-power by omitting the necessity of additional fat suppression pulses.

Appendix E

Weighted Averaging – Phase Encoding Oversampling

Weighted averaging was introduced by M.von Kienlin[10] as a means to improve the SNR of 31P-CSI. It is based on the insight, that signal intensity of k-space data around the k-space center is higher compared to the k-space periphery. If predominantly such data are used for averaging, SNR will increase faster than for isotropic averaging.

Fig.E1 a) shows a typical averaging scheme. A full acquisition(red) is followed by acquisition of the center half(blue) and the central quarter of k-space. Realizing the sampling scheme consecutively with segmented acquisition leads to the sampling scheme shown in b) and c). b) shows the expanded sampling scheme with interleaved sampling for each average. The phase encoding order along the echo trains is shown in c). It demonstrates, that in this concatenated sampling identical phase encoding steps are acquired at different echo times, which may lead to artifacts. A more continuous sampling scheme is shown in d): the phase encoding steps are first combined and sorted. Interleaving (e) then leads to a more continuous sampling pattern, identical phase encoding steps are sampled at consistent (or identical) echo

Fig.E1

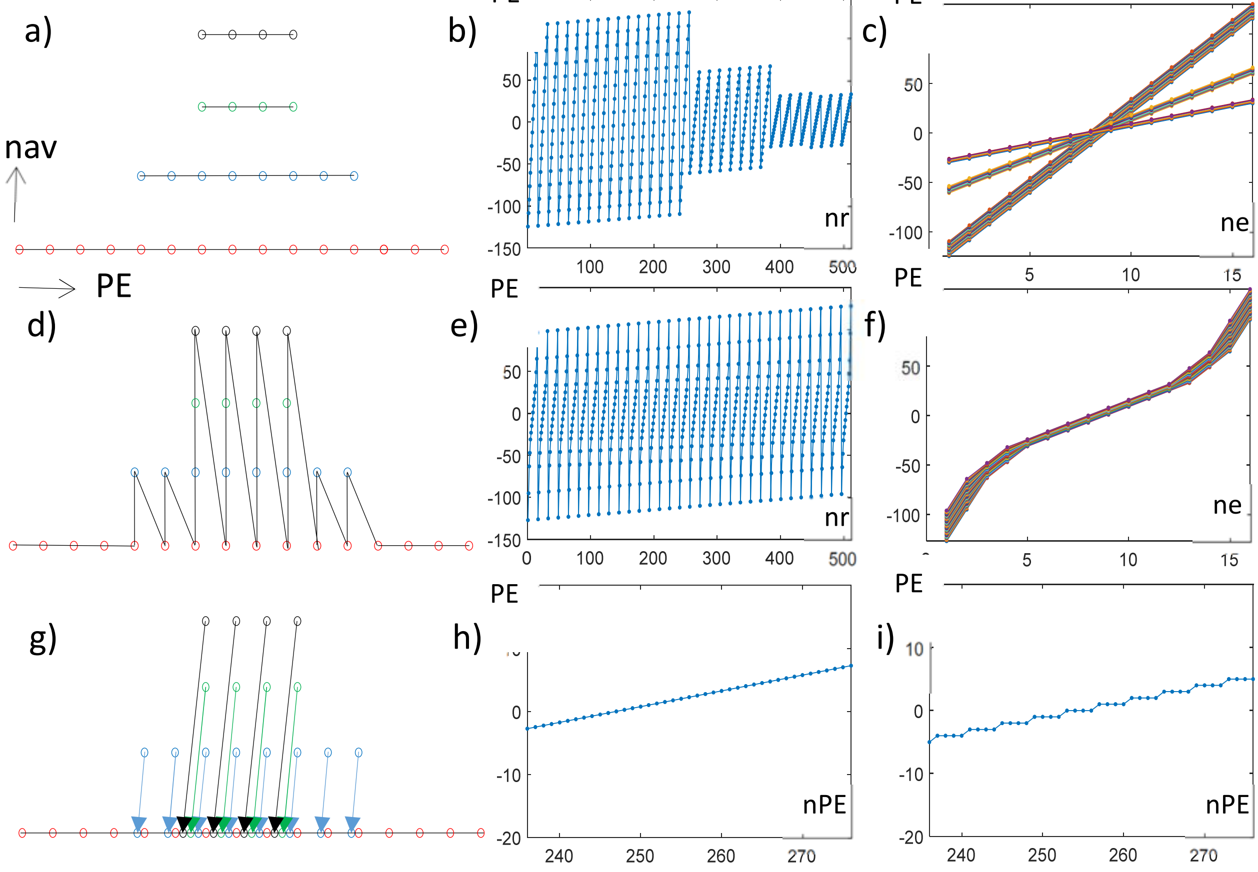


Fig.E1 a) basic sampling scheme of weighted averaging: In higher number of averages nav only data around the k-space center are acquired. B) PE-steps along the number of readouts nr for concatenated averaging. C) PE-steps along the echotrain. d)-e) as a)-c) for continuous averaging. g) projecting the higher averages with an increasing shift by fradtions of the PE step width leads to phase encoding oversampling. H) shows the central part of sorted phase encoding order of phase encoding oversampling, i) the same for the continuous scheme (d), in which identical phase encoding steps are repeated times (f). In a further modification the PE-steps are incremented by one quarter of the PE-step width. This then leads to a very similar sampling pattern as in e) and f), but after re-sorting k-space is covered monotonously (h) compared to a stepwise coverage in d). In the end the sampling scheme shown in g) leads to phase encoding undersampling, which allows inner volume imaging in the phase encoding direction.

Fig.E2 shows an example of the results of imaging with low SNR (low field, thin slice). a) corresponds to a single acquisition, b) to two conventional averages, c) to weighted averaging with the weighting scheme [1 0.5 0.25 0.25] and continuous acquisition (acqP.PEmode='cont'), which leads to the same number of acquisition steps as in b). The improvement in image quality is clearly demonstrated. A compromise between SNR and blurring can be made by weighting each PE-step with a weighting function

$$W(PE)=nav(PE)^{-k}$$

Where k is varied between 0… 1. For k=0 no weighting is performed and blurring becomes more pronounced, for k=1 no blurring occurs, but SNR is only mildly improved. Images in c) where reconstructed with k=1/2. The proper choice of k is determined by the SNR of the single scan.

Fig.E2

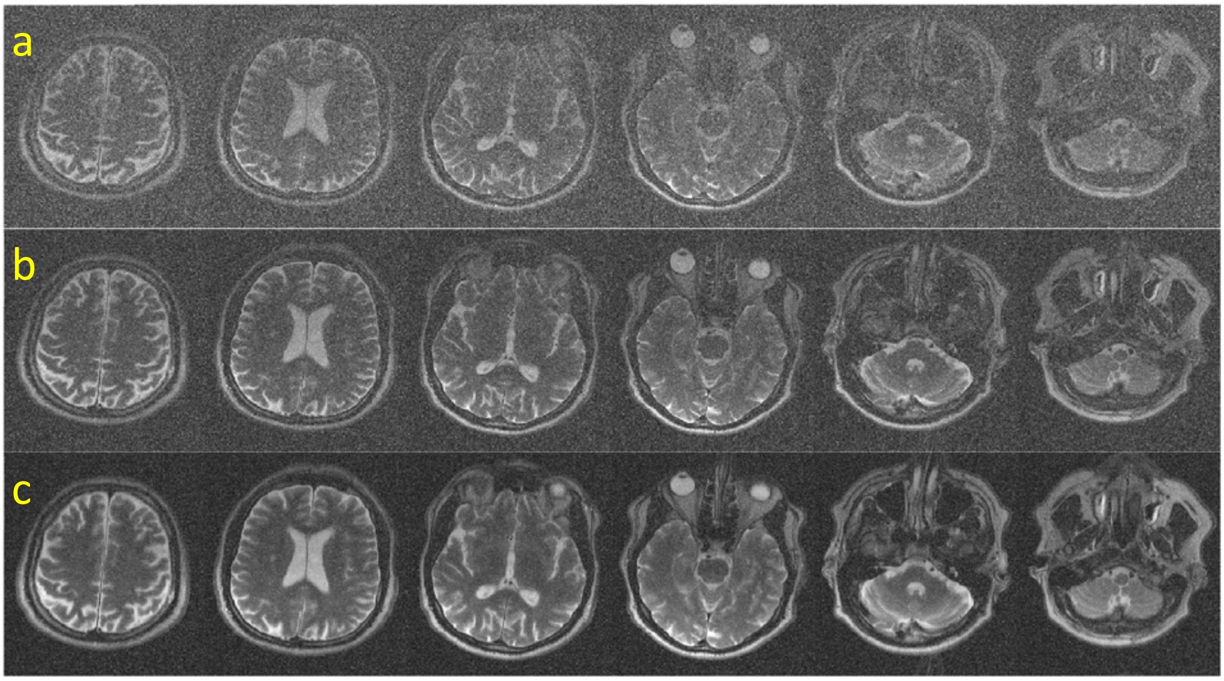


Fig.E2 results of weighted averaging

Weighted averaging can of course be combined with more advanced AI-based image reconstruction for further improvement of SNR. A typical case are scans with very low SNR, where AI-reconstruction alone would lead to increased risk of 'hallucinations'.

Good candidate scenarios for weighted averaging are imaging of heteronuclei (Na, 35P, 13C) and/or chemical shift imaging as in the original publication.

The application scenario of phase encoding oversampling is more general and not only applies to low SNR scans. Th phase encoding undersampling allows true inner volume imaging like high-resolution imaging of inner organs and body parts like the prostate, pancreas, eyes, inner ears, A sampling scheme with acqP.PEinc=0.5 and acqP:wav=[1 0.1] corresponds to a GRAPPA-acquisition with phase encoding oversampling by a factor of 2, where 10% of the acquired data represent the reference data for the GRAPPA-kernel.

Weighted averaging can be combined with acceleration schemes like half fourier imaging (acqP.HF_fac>0) or GRAPPA (acqP.GR_fac>1). For improving SNR this is somewhat questionable because they lead to an a priori reduction of SNR, but for inner volume imaging with phase encoding oversampling this may be a valid approach.

Appendix F

Variable Density Sampling

Variable Density Sampling was initially developed to allow non isotropic sampling of k-space data in the readout direction similar to weighted averaging in the phase encoding direction. This is achieved by using a V-shaped readout gradient during the acquisition time acq (s.Fig.F1c). The depth of the V is determined by the parameter acqP.GRramp. acqP.GRramp is scaled in units of the amplitude of the readout gradient. acqP.GRramp=1 sets the readout gradient at the center of kx to zero. The V is placed symmetrically around the readout gradient, the area under the gradient and thus the spatial resolution remains unchanged. As seen in comparison of Fig.F1b) and d) the sampled k-space remains the same but with a more dense sampling around kx=0.

Fig.F1

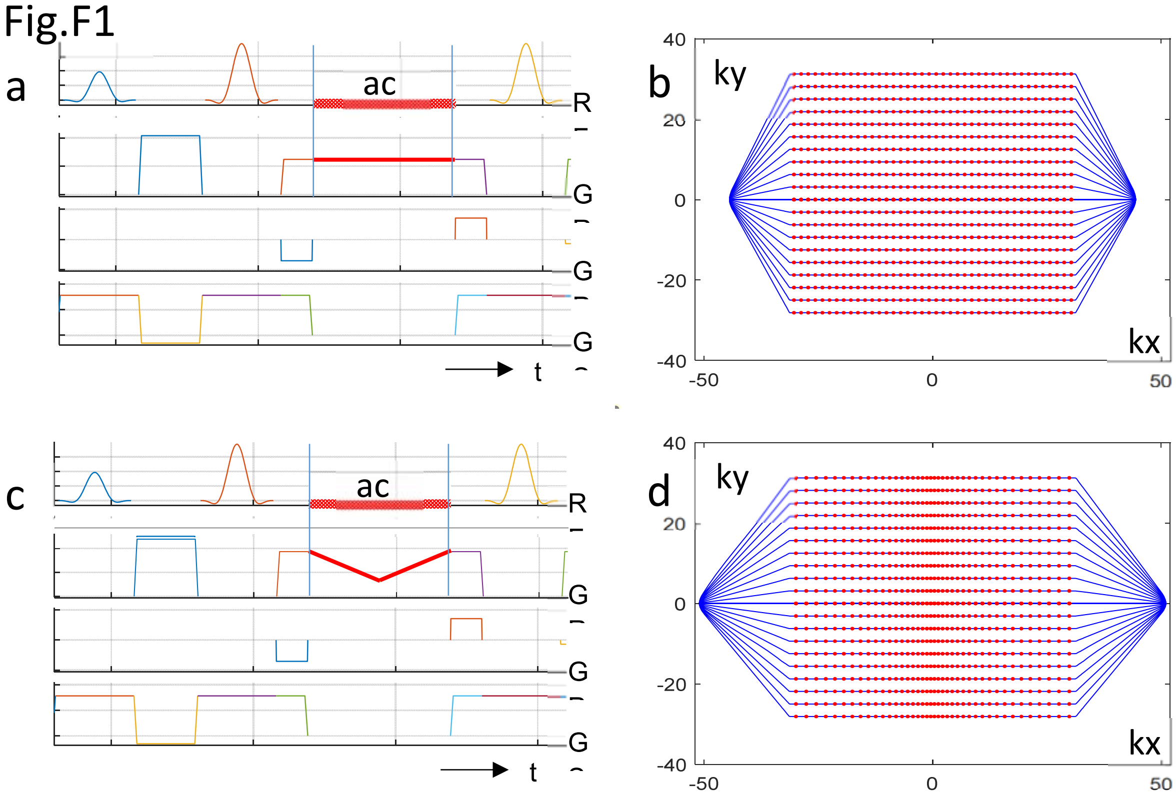


Fig.F1 Sequence diagram for rectilinear (a) and variable density (c) sampling and the corresponding k-space coverage (b, d).

Lower readout gradient at the center of kx means lower acquisition bandwidth and therefore higher SNR. The shape of the k-space trajectory remains unchanged, but non-uniform Fourier transform is required to reconstruct the images due to the variable sampling density.

Fig.F2 shows results of measurements with different values of acqP.GRramp ranging from 0,0.2,0.4,0.6,0.8 in (a)…(e). Applying FFT to the variable density data leads to increasing distortions in the readout-direction. No distortions but increasing blurring is observed with nuFFT-reconstruction. The zoomed images of the inner ear show, that images appear sharp up to acqP.GRramp ~ 0.3-0.4. The profile at the position of the yellow line demonstrate the increase in SNR with increasing acqP.GRramp.

The variable distance between sampling points leads to a variable field-of-view, which becomes increasingly small as one goes out to higher kx-coordinates. In order to avoid foldover artifacts it is advisable to increase the oversampling factor acq.accfac.

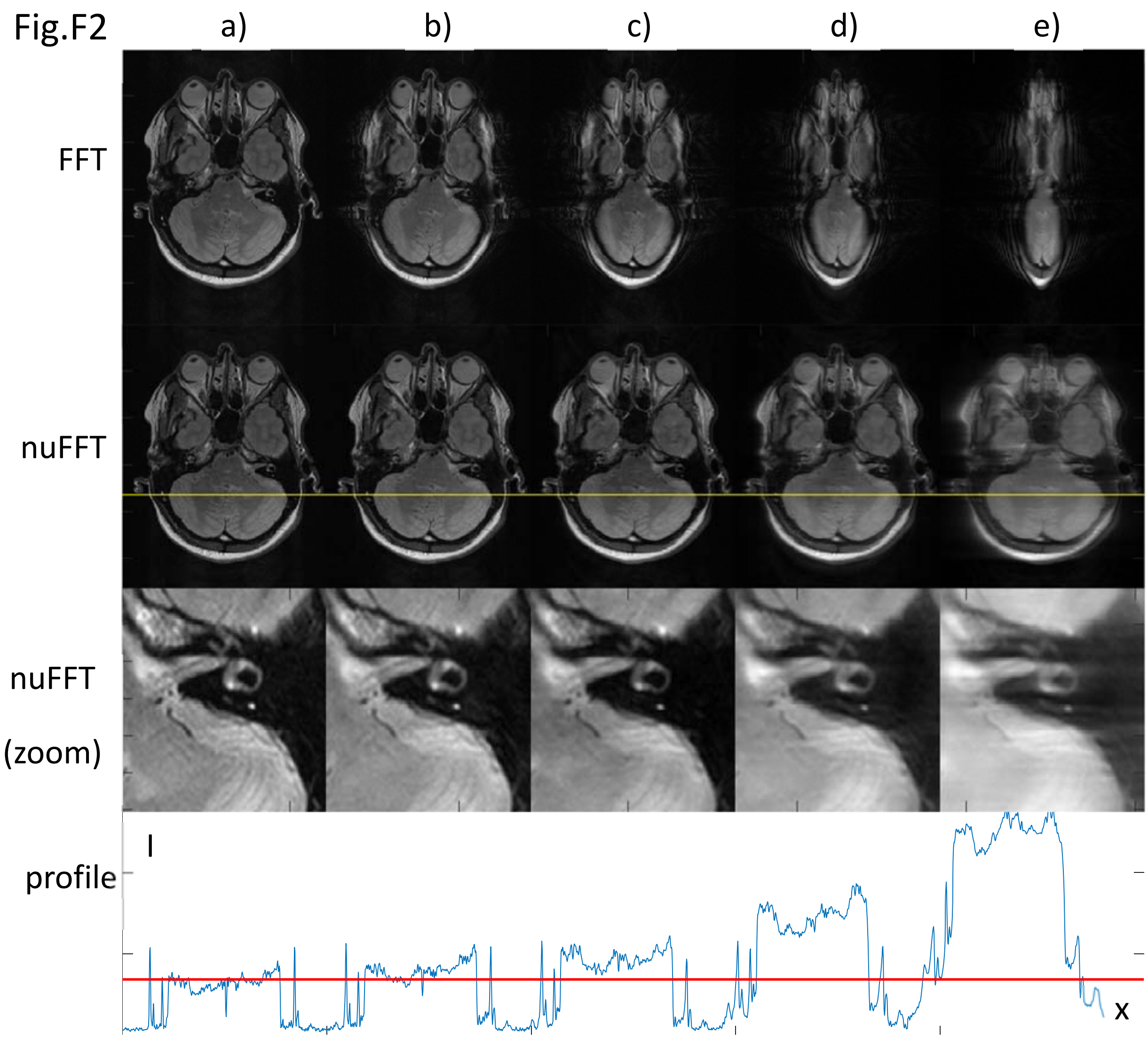


Fig.F2 images acquired with different values of acqP.GRramp. For explanation see text.

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